



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE – (24AD3107)

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TITLE OF THE PROJECT	Interactive Exploratory Data Analysis Dashboard	
S. No.	University ID	Name
1	2420030665	M.VAISHNAVI
2	2420030230	V.VARSHITA
3	2420030120	SHAIK ALIZA
Name of the Guide		PRATUSHA MAM

Abstract:

Exploratory Data Analysis (EDA) is an important step in the data analysis process because it helps users understand the structure, quality, patterns, and relationships present in a dataset. With the increasing amount of data generated from different fields such as education, business, healthcare, finance, and research, analyzing datasets efficiently has become essential. However, performing EDA manually can be time-consuming and usually requires programming knowledge. Users often need to write separate code for loading data, calculating statistics, identifying missing values, detecting outliers, and creating different visualizations. To overcome these limitations, this project proposes an Interactive EDA Dashboard for Automated CSV Data Analysis that provides a simple and interactive platform for performing common exploratory data analysis tasks without requiring users to write code.

The proposed system allows users to upload a CSV file through a web-based dashboard. After uploading the dataset, the system automatically reads and processes the data and provides important information about the dataset. This includes the number of rows and columns, column names, data types, missing values, duplicate records, and descriptive statistics. The system can identify numerical and categorical columns and provide suitable analysis based on the type of

data. This helps users quickly understand the overall structure and quality of their dataset before performing further analysis.

One of the main features of the proposed system is the automatic generation of different data visualizations. Distribution plots can be generated to understand the frequency, spread, and distribution of numerical variables. Box plots can be used to identify the median, variation, and possible outliers in the dataset. A correlation heatmap can show the relationship between different numerical variables and help users identify highly correlated features. Pair plots can also be generated to visualize relationships between multiple numerical variables and identify patterns, trends, clusters, and unusual observations. These visualizations provide users with a clear graphical representation of the dataset and make complex data easier to understand.

The dashboard will be developed using Python and Streamlit. Pandas will be used for loading, processing, and analyzing CSV data, while NumPy will support numerical operations. Matplotlib and Seaborn will be used to generate statistical graphs and visualizations. Streamlit will provide the interactive web interface through which users can upload datasets, select variables, and view the generated analysis. The combination of these technologies makes it possible to develop a lightweight, flexible, and user-friendly EDA platform.

The proposed dashboard aims to reduce the repetitive effort involved in performing basic exploratory data analysis. It can be useful for students, researchers, beginners, and data analysts who want to understand a dataset quickly without writing separate programs for every analysis. By providing statistical information and multiple visualizations in a single platform, the system can make the initial data analysis process faster and more accessible. The project can also act as a foundation for further data science and machine learning tasks, as understanding the dataset is an important step before preprocessing, feature selection, and model development.

In the future, the system can be enhanced with automated data cleaning, advanced outlier detection, categorical data visualizations, downloadable EDA reports, support for additional file formats, and interactive charts. Artificial intelligence and machine learning techniques can also be integrated to generate automatic insights and recommendations based on the uploaded dataset. Thus, the Interactive EDA Dashboard provides an efficient and accessible approach to automated exploratory data analysis by combining data processing, statistical analysis, and visualization in a single interactive platform.

Project Architecture/Project Overview:

